is the process of fertilization. Let us recall the sexual reproduction in algae and bryophytes. They reproduce by the production of gametes which may be motile or non motile depending upon the species. The gametic fusion is of three types (Isogamy, Anisogamy and Oogamy). In algae external fertilization takes place whereas in higher plants internal fertilization occurs.

Flower

A flower is viewed in multidimensional perspectives from time immemorial. It is an inspirational tool for the poets. It is a decorative material for all the celebrations. In Tamil literature the five lands are denoted by different flowers. The flags of some countries are embedded with flowers. Flowers are used in the preparation of perfumes. For a Morphologist, a flower is a highly condensed shoot meant for reproduction. As you have already learned about the parts of a flower in Unit II of Class XI, let us recall the parts of a flower. A Flower possesses four whorls- Calyx, Corolla, Androecium and Gynoecium. Androecium and Gynoecium are essential organs(Figure 1.3). The process or changes involved in sexual reproduction of higher plants include three stages .They are Pre-fertilization, Fertilization and Post fertilization changes. Let us discuss these events in detail.

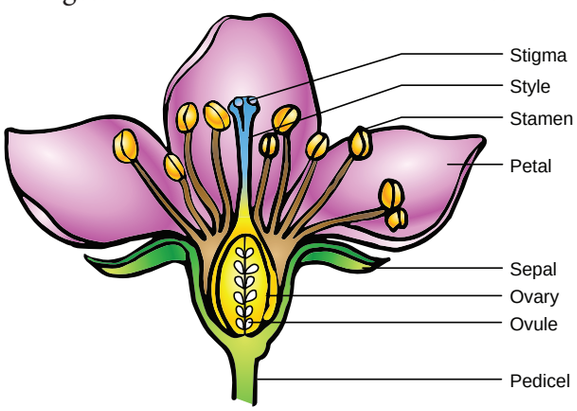


Figure 1.3 Parts of a Flower

1.4 Pre-fertilization: structures and events

The hormonal and structural changes in plant lead to the differentiation and development of floral primordium. The structures and events involved in pre-fertilization are given below

1.4.1 Male Reproductive part - Androecium

Androecium is made up of stamens. Each stamen possesses an anther and a filament. Anther bears pollen grains which represent the male gametophyte. In this chapter we shall discuss the structure and development of anther in detail.

Development of anther: A very young anther develops as a homogenous mass of cells surrounded by an epidermis. During its development, the anther assumes a four-lobed structure. In each lobe, a row or a few rows of hypodermal cells becomes enlarged with conspicuous nuclei. This functions as archesporium. The archesporial cells divide by periclinal divisions to form primary parietal cells towards the epidermis and primary sporogenous cells towards the inner side of the anther. The primary parietal cells undergo a series of periclinal and anticlinal division and form 2-5 layers of anther walls composed of endothecium, middle layers and tapetum, from periphery to centre.

Microsporogenesis: The stages involved in the formation of haploid microspores from diploid microspore mother cell through meiosis is called **Microsporogenesis**. The primary sporogeneous cells directly, or may undergo a few mitotic divisions to form **sporogenous tissue**. The last generation of sporogenous tissue functions as microspore mother cells. Each microspore mother cell divides meiotically to form a tetrad of four haploid microspores (microspore tetrad). Microspores soon separate from one another and remain free in the anther locule and develop into pollen grains. The stages in the development of microsporangia is given in Figure 1.4. In some plants, all the microspores in a microsporangium remain held together called **pollinium**. Example: *Calotropis*. Pollinia are

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